

Plant competition as a mechanism of invasion on islands: Revisiting the conclusions of Kuebbing and Nuñez (2016)

Kasey E. Barton  | Amanda Wong

Department of Botany, University of Hawai'i at Mānoa, Honolulu, Hawai'i

Correspondence

Kasey E. Barton, Department of Botany, University of Hawai'i at Mānoa, 3190 Maile Way, Honolulu, Hawai'i 96822.
Email: kbarton@hawaii.edu

Associate Editor: Orou Gaoue
Handling Editor: Emilio Bruna

Abstract

Identifying the mechanisms underlying invasive plant establishment and native plant decline remains a central goal in ecology, particularly for biodiversity hotspots such as islands. We re-analyzed a previously published meta-dataset to test the prediction that neighbor effects are stronger on islands vs. continents because island plants are weaker competitors. Although we detected marginally stronger neighbor effects on islands than continents, this was due, at least in part, to stronger competition among native species. An absence of studies testing neighbor effects between native vs. invasive plants on islands prevents a rigorous test for competition effects on islands, highlighting an important knowledge gap.

KEYWORDS

island rules, meta-analysis, neighbor effects, plant invasions

Invasive plants disrupt native plant community dynamics, ecosystem function, agriculture, and forestry, globally costing billions of dollars in management each year (Pimentel, Zuniga & Morrison, 2005). Identifying the mechanisms underlying invasive plant establishment and native plant decline remains a central goal in ecology. A recent meta-analysis highlighted the widespread evidence that plant–plant interactions are involved in plant invasion dynamics (Kuebbing & Nunez, 2016). In particular, it was shown that under experimental conditions, native plants were more suppressed in growth than non-native plants when grown with non-native neighbors. Moreover, native neighbors tended not to suppress the growth of non-native focal plants at all (effect size overlaps with zero). These results suggest that native plants are weak competitors compared to non-native plants, suggesting that invasive plants displace native plants by outcompeting them.

For biodiversity hotspots, such as insular islands, plant invasions are particularly devastating and are the primary drivers of native plant declines (Caujape-Castells et al., 2010). Identifying mechanisms of plant invasions on islands will not only improve management efforts by targeting the most critical processes of invasive

species success, but it can also shed light on predicted inherent differences between island and continental plants, the so-called “island rules” (Patino et al., 2017). For example, low species richness on islands could result in the availability of niche space, which when filled by fast growing and vigorous invasive species could lead to the displacement of native plants that are outcompeted by invasive species (Simberloff, 1995; Stachowicz & Tilman, 2005). To assess whether competition effects of non-native on native plants are generally stronger on islands than continents, we re-analyzed the Kuebbing and Nunez meta-dataset after adding the additional variable of site (island vs. continent).

Searches on Web of Science (using the same keywords as Kuebbing & Nunez, 2016 with the addition of “island”) revealed no additional studies testing plant competition on islands published since 2016. In addition, we perused all publications that have cited Kuebbing and Nunez (2016), none of which provided new experimental tests of competition on islands. Thus, our new analyses were conducted on the meta-dataset published by Kuebbing and Nunez (2016). After reviewing the full meta-dataset, we determined that 10 of the 83 publications were conducted on islands (Britton,

Marrs, Pakeman & Carey, 2003; Brooker, Scott, Palmer & Swaine, 2006; D'Antonio, Hughes & Vitousek, 2001; Garbuzov, Reidinger & Hartley, 2011; van Mierlo, Wilms & Berendse, 2000; Stevens & Gowing, 2014; Suzuki & Suzuki, 2011a,b; Takakura & Fujii, 2010; del-Val & Crawley, 2004), yielding a total of $n=62$ pairwise comparisons of neighbor effects on islands. The islands represented in this dataset are as follows: Great Britain, New Zealand, Japan, and Hawaii; studies were conducted in both field and greenhouse experiments; and species represented are largely perennial herbs and grasses with few woody plants assessed.

We conducted a meta-analysis to test whether neighbor effects differed between islands and continents using the package “metafor” (Viechtbauer, 2014) package in R version 3.4.3. The effect size used was Hedges’ d , which includes a correction factor for small sample sizes and is robust to unequal sampling variances (Gurevitch, Curtis & Jones, 2001). The significance of site (island vs. continent) was tested in mixed effects models using the `rma.mv()` function, including publication as a random factor. Model-generated group means and 95% confidence intervals are presented in Figure 1; significant effect sizes are those with confidence intervals that do not overlap with zero. This analysis revealed that as predicted, neighbor effects are marginally stronger on islands than continents ($Q_M = 3.3116$, $df = 1$, $p = 0.0688$; Figure 1). However, this analysis fails to specifically test whether island plants are more vulnerable to invasive plant competition than continental plants because none of the studies included tested native vs. non-native neighbor effects.

Additional insights were gained by partitioning the dataset to separately assess neighbor effects of native vs. native and non-native vs. non-native plants. We detected significantly stronger neighbor

effects between native species on islands compared to continents ($Q_M = 5.4490$, $df = 1$, $p = 0.0196$; Figure 1), suggesting that in contrast to the island rule prediction, competitive interactions may be stronger among native island than continental plants. With the exception of New Zealand, the island floras represented in this analysis are comparable to temperate forests with relatively low species richness. Only Hawaii is sub-tropical, and it has much lower species richness than other sub-tropical and tropical forests (Ostertag, Inman-Narahari, Cordell, Giardina & Sack, 2014). Strong competition among native island plants could evolve due to the high likelihood of species co-occurring in the relatively low-richness island floras (Gilbert & Lechowicz, 2004; Hubbell, 2005). In contrast, in high-diversity forests, neutral theory has posited that community assembly via competitive niche partitioning is unlikely given the requirement of equilibrium, which is effectively never reached in time (Hubbell, 2005). Moreover, if competitive interactions do not consistently occur between species over evolutionary time due to the relatively low abundances within species, competitive niche partitioning may never arise in high-diversity forests (Hubbell, 2001). No difference was detected in the neighbor effects of non-native plants on other non-native plants on islands vs. continents ($Q_M = 0.0087$, $df = 1$, $p = 0.9258$; Figure 1), indicating that invasive species do not appear to be competing more aggressively against other invasive species on islands than continents.

This meta-analysis, which capitalizes on a previously published dataset (Kuebbing & Nunez, 2016), highlights important gaps in our knowledge of the role of competition in plant invasions on islands. Despite the primary role invasive plants play in the decline of native island floras (Caujape-Castells et al., 2010), there is a paucity of experimental studies directly testing competition between native vs. invasive island plants (D'Antonio, Hughes, Mack, Hitchcock & Vitousek, 1998). Although community-scale studies report patterns consistent with competition between invasive and native plants on islands (Cordell et al., 2009; Monty, Florens & Baider, 2013; Walker & Vitousek, 1991; Yessoufou, Bezeng, Gaoue, Bengu & van der Bank, 2018), there is an urgent need for studies designed explicitly to test competition. Finally, the research bias of primarily studying islands in temperate regions (e.g., Japan, Great Britain, New Zealand) means that the role of competition in plant invasions remains particularly unclear in tropical islands. High geographic isolation and endemism make tropical islands particularly vulnerable to plant invasions (Caujape-Castells et al., 2010), highlighting the importance of identifying mechanisms of invasion for conservation of these biodiversity hotspots.

ACKNOWLEDGMENTS

The authors thank K. Feliciano, A. Westerband and T. Lum for discussions contributing to this meta-analysis. Feedback from two reviewers improved the manuscript.

DATA ACCESSIBILITY

Data available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.rc7bh42> (Barton & Wong, 2019).

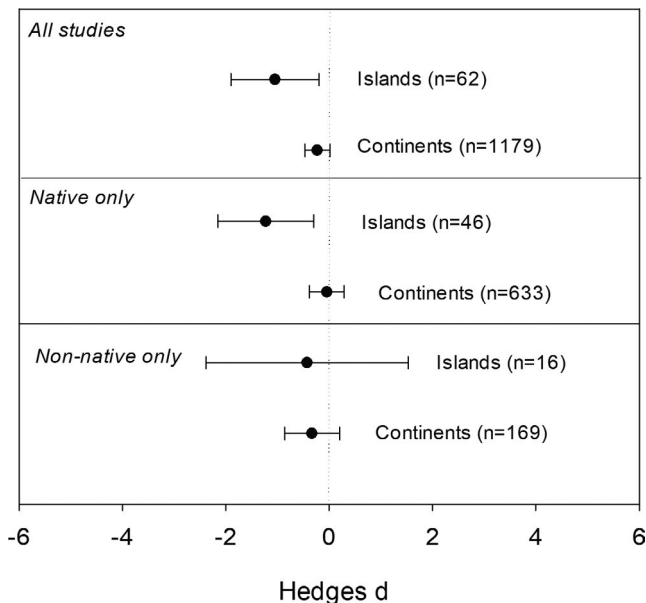


FIGURE 1 Mean Hedges d effect sizes with bias-corrected 95% confidence intervals for experimental effects of neighboring plants in island vs. continental studies. Comparisons are shown for the entire data set (Kuebbing & Nunez, 2016), as well as for subsets of native vs. native and non-native vs. non-native neighbor trials. The number of effect sizes is reported in parentheses

ORCID

Kasey E. Barton  <https://orcid.org/0000-0002-4549-4150>

REFERENCES

- Barton, K. E., & Wong, A. (2019). Data from: Plant competition as a mechanism of invasion on islands: Revisiting the conclusions of Kuebbing and Nuñez (2016). *Dryad Digital Repository*. <https://doi.org/10.5061/dryad.rc7bh42>
- Britton, A., Marrs, R., Pakeman, R., & Carey, P. (2003). The influence of soil-type, drought and nitrogen addition on interactions between *Calluna vulgaris* and *Deschampsia flexuosa*: Implications for heathland regeneration. *Plant Ecology*, 166, 93–105. <https://doi.org/10.1023/A:1023248407408>
- Brooker, R. W., Scott, D., Palmer, S. C. F., & Swaine, E. (2006). Transient facilitative effects of heather on Scots pine along a grazing disturbance gradient in Scottish moorland. *Journal of Ecology*, 94, 637–645. <https://doi.org/10.1111/j.1365-2745.2006.01129.x>
- Caujape-Castells, J., Tye, A., Crawford, D. J., Santos-Guerra, A., Sakai, A., Beaver, K., ... Kueffer, C. (2010). Conservation of oceanic island floras: Present and future global challenges. *Perspectives in Plant Ecology Evolution and Systematics*, 12, 107–129. <https://doi.org/10.1016/j.ppees.2009.10.001>
- Cordell, S., Ostertag, R., Rowe, B., Schweinhart, L., Vasquez-Radonic, L., Michaud, J., ... Schulten, J. R. (2009). Evaluating barriers to native seedling establishment in an invaded Hawaiian lowland wet forest. *Biological Conservation*, 142, 2997–3004. <https://doi.org/10.1016/j.biocon.2009.07.033>
- D'Antonio, C. M., Hughes, R. F., Mack, M., Hitchcock, D., & Vitousek, P. M. (1998). The response of native species to removal of invasive exotic grasses in a seasonally dry Hawaiian woodland. *Journal of Vegetation Science*, 9, 699–712. <https://doi.org/10.2307/3237288>
- D'Antonio, C. M., Hughes, R. F., & Vitousek, P. M. (2001). Factors influencing dynamics of two invasive C-4 grasses in seasonally dry Hawaiian woodlands. *Ecology*, 82, 89–104.
- del-Val, E., & Crawley, M. J. (2004). Interspecific competition and tolerance to defoliation in four grassland species. *Canadian Journal of Botany-Revue Canadienne De Botanique*, 82, 871–877.
- Garbuzov, M., Reidinger, S., & Hartley, S. E. (2011). Interactive effects of plant-available soil silicon and herbivory on competition between two grass species. *Annals of Botany*, 108, 1355–1363. <https://doi.org/10.1093/aob/mcr230>
- Gilbert, B., & Lechowicz, M. J. (2004). Neutrality, niches, and dispersal in a temperate forest understory. *Proceedings of the National Academy of Sciences of the United States of America*, 101, 7651–7656. <https://doi.org/10.1073/pnas.0400814101>
- Gurevitch, J., Curtis, P. S., & Jones, M. H. (2001). Meta-analysis in ecology. *Advances in Ecological Research*, 32, 199–247. [https://doi.org/10.1016/S0065-2504\(01\)32013-5](https://doi.org/10.1016/S0065-2504(01)32013-5)
- Hubbell, S. P. (2001). *The unified neutral theory of biodiversity and biogeography*. Princeton, NJ: Princeton University Press.
- Hubbell, S. P. (2005). Neutral theory in community ecology and the hypothesis of functional equivalence. *Functional Ecology*, 19, 166–172. <https://doi.org/10.1111/j.0269-8463.2005.00965.x>
- Kuebbing, S. E., & Nunez, M. A. (2016). Invasive non-native plants have a greater effect on neighbouring natives than other non-natives. *Nature Plants*, 2, 7.
- Monty, M. L. F., Florens, F. B. V., & Baider, C. (2013). Invasive alien plants elicit reduced production of flowers and fruits in various native forest species on the tropical island of Mauritius (Mascarenes, Indian Ocean). *Tropical Conservation Science*, 6, 35–49. <https://doi.org/10.1177/194008291300600107>
- Ostertag, R., Inman-Narahari, F., Cordell, S., Giardina, C. P., & Sack, L. (2014). Forest structure in low-diversity tropical forests: A study of Hawaiian wet and dry forests. *PLoS ONE*, 9, e103268. <https://doi.org/10.1371/journal.pone.0103268>
- Patino, J., Whittaker, R. J., Borges, P. A. V., Fernandez-Palacios, J. M., Ah-Peng, C., Araujo, M. B., ... Emerson, B. C. (2017). A roadmap for island biology: 50 fundamental questions after 50 years of The Theory of Island Biogeography. *Journal of Biogeography*, 44, 963–983. <https://doi.org/10.1111/jbi.12986>
- Pimentel, D., Zuniga, R., & Morrison, D. (2005). Update on the environmental and economic costs associated with alien-invasive species in the United States. *Ecological Economics*, 52, 273–288. <https://doi.org/10.1016/j.ecolecon.2004.10.002>
- Simberloff, D. (1995). Why do introduced species appear to devastate islands more than mainland areas? *Pacific Science*, 49, 87–97.
- Stachowicz, J., & Tilman, D. (2005). Species invasions and the relationships between species diversity, community saturation, and ecosystem function. In D. F. Sax, J. J. Stachowicz, & S. D. Gaines (Eds.), *Species invasions: Insights into ecology, evolution, and biogeography* (pp. 41–64). Sunderland, MA: Sinauer Associates.
- Stevens, C. J., & Gowing, D. J. G. (2014). Effect of nitrogen addition, form and clipping on competitive interactions between grassland species. *Journal of Plant Ecology*, 7, 222–230. <https://doi.org/10.1093/jpe/rtt039>
- Suzuki, R. O., & Suzuki, S. N. (2011a). Facilitative and competitive effects of a large species with defensive traits on a grazing-adapted, small species in a long-term deer grazing habitat. *Plant Ecology*, 212, 343–351. <https://doi.org/10.1007/s11258-010-9826-6>
- Suzuki, S. N., & Suzuki, R. O. (2011b). Distance-dependent shifts in net effects by an unpalatable nettle on a palatable plant species. *Acta Oecologica-International Journal of Ecology*, 37, 386–392. <https://doi.org/10.1016/j.actao.2011.04.011>
- Takakura, K. I., & Fujii, S. (2010). Reproductive interference and salinity tolerance differentiate habitat use between two alien cockleburs: *Xanthium occidentale* and *X. italicum* (Compositae). *Plant Ecology*, 206, 309–319. <https://doi.org/10.1007/s11258-009-9644-x>
- van Mierlo, J. E. M., Wilms, Y. J. C., & Berendse, F. (2000). Effects of soil organic matter and nitrogen supply on competition between *Festuca ovina* and *Deschampsia flexuosa* during inland dune succession. *Plant Ecology*, 148, 51–59. <https://doi.org/10.1023/A:1009852630716>
- Viechtbauer, W. (2014). *Meta-analysis Package for R*.
- Walker, L. R., & Vitousek, P. M. (1991). An invader alters germination and growth of a native dominant tree in Hawaii. *Ecology*, 72, 1449–1455. <https://doi.org/10.2307/1941117>
- Yessoufou, K., Bezeng, B. S., Gao, O. G., Bengu, T., & van der Bank, M. (2018). Phylogenetically diverse native systems are more resistant to invasive plant species on Robben Island, South Africa. *Genome*, 1–12. <https://doi.org/10.1139/gen-2018-0039>

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Barton KE, Wong A. Plant competition as a mechanism of invasion on islands: Revisiting the conclusions of Kuebbing and Nuñez (2016). *Biotropica*. 2019;51:316–318. <https://doi.org/10.1111/btp.12659>